Cart lab: change driving force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use controlled data to describe how net force affects acceleration.

SUCCESS CRITERIA - I can...

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “net force” to explain your model or result. Sentence starters are optional.

constant: not changing during the stated interval

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

01 Bring back an earlier idea

A puck follows a circle at a steady 6 m/s. A student calls it equilibrium.
Draw velocity at two points and correct the claim.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a = F_{net}/M$; $F_{drive} = m_h g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

Worked reasoning: A 2 kg cart has net forces 2 and 6 N in two trials. Using a=F/m gives accelerations 1 and 3 m/s². Tripling net force triples acceleration.

Your turn: Draw a force-versus-acceleration graph for these trials.
Explain the meaning of the slope.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

03 Interpret a force graph

For total mass 1.00 kg, use the ideal model table to find the slope of a versus net force.
State its units and recover the mass.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.

F_{net} (N)	a(m/s²)
0.245	0.245
0.490	0.490
0.735	0.735
0.980	0.980

Marzano check - circle: ① ② ③

04 Design the controlled trial

Explain how transferring masses from cart to hanger changes driving weight without changing total moving mass. Plan a table with three force settings and three acceleration trials each.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Both variables change

System A has net force 2 N and mass 1 kg. B has 6 N and 2 kg.
Compare accelerations and explain why tripling force does not triple acceleration here.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A 0.3 kg system accelerates at 0.5 m/s².
Predict acceleration for 0.7 kg at the same net force.
Explain the effect of greater resistance.

Word help: acceleration = rate of change of velocity

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

An a-versus-F_{net} graph has slope 2.0(m/s²)/N.
Find mass.
Predict a at 1.5 N.
Explain why plotting driving force instead of net force can produce an offset.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2 kg cart is pulled with 3, 5, 7 N and has accelerations 1, 2, 3 m/s². Infer a constant opposing force and show how the graph reveals it.

Word help: infer = use evidence to reach a conclusion

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3